

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (Bsc.IT Semester III)
Data Structures Practical

Practical - VIII

Roll No. : S057	Name: Amit Vishwakarma
Class : SY IT	Batch : 1
Date of Assignment : 12/08/26	Date/Time of Submission : 12/08/26

Sorting Algorithms:

Write programs to implement and compare the following sorting algorithms:

- a. Bubble sort

Code:

```
# Bubble sort in Python
def bubbleSort(array):

    # loop to access each array element
    for i in range(len(array)):

        # loop to compare array elements
        for j in range(0, len(array) - i - 1):

            # compare two adjacent elements
            # change > to < to sort in descending order
            if array[j] > array[j + 1]:

                # swapping elements if elements
                # are not in the intended order
                temp = array[j]
                array[j] = array[j+1]
                array[j+1] = temp

data = [-2, 45, 0, 11, -9]
bubbleSort(data)

print('Sorted Array in Ascending Order:')
print(data)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/Users/mvlui/One
Sorted Array in Ascending Order:
[-9, -2, 0, 11, 45]

Amit_S057
|
```

b. Insertion sort

Code:

```
# Selection sort in Python
def selectionSort(array, size):
    for step in range(size):
        min_idx = step

        for i in range(step + 1, size):

            # to sort in descending order, change > to < in this line
            # select the minimum element in each loop
            if array[i] < array[min_idx]:
                min_idx = i

        # put min at the correct position
        (array[step], array[min_idx]) = (array[min_idx], array[step])

data = [-2, 45, 0, 11, -9]
size = len(data)
selectionSort(data, size)
print('Sorted Array in Ascending Order:')
print(data)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/Users/m
Sorted Array in Ascending Order:
[-9, -2, 11, 0, 45]

Amit_S057
|
```

c. Selection sort

Code:

```
# Insertion sort in Python
def insertionSort(array):
    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        # Compare key with each element on the left of it until an element smaller than it is found
        # For descending order, change key<array[j] to key>array[j].
        while j >= 0 and key < array[j]:
            array[j + 1] = array[j]
            j = j - 1

        # Place key at after the element just smaller than it.
        array[j + 1] = key

data = [9, 7, 1, 5, 2]
insertionSort(data)
print('Sorted Array in Ascending Order:')
print(data)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/Users/mv
Sorted Array in Ascending Order:
[1, 2, 5, 7, 9]

Amit_S057
```

Curiosity Questions:

Programming Questions

1. Write a Python program to sort a list of numbers in descending order using **Bubble Sort**.

Code:

```
def bubbleSort(array):
    n = len(array)

    for i in range(n):
        for j in range(0, n - i - 1):

            # Descending order
            if array[j] < array[j + 1]:
                array[j], array[j + 1] = array[j + 1], array[j]

data = [7, 2, 10, 6, 1, 8]

bubbleSort(data)

print("Sorted Array in Descending Order:")
print(data)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/Users/mvl
Sorted Array in Descending Order:
[10, 8, 7, 6, 2, 1]

Amit_S057
|
```

2. Write a Python program to sort a list of numbers in descending order using **Insertion Sort**.

Code:

```
def insertionSort(array):
    for step in range(1, len(array)):
        key = array[step]
        j = step - 1

        # Descending order
        while j >= 0 and key > array[j]:
            array[j + 1] = array[j]
            j = j - 1

        array[j + 1] = key

data = [9, 5, 1, 4, 3]
insertionSort(data)

print('Sorted Array in Descending Order:')
print(data)
print("\nAmit_S057")
```

Output:

```
===== RESTART: C:/Users/mv|
Sorted Array in Descending Order:
[9, 5, 4, 3, 1]

Amit_S057
```

3. Write a Python program to sort a list of numbers in descending order using **Selection Sort**.

Code:

```
def selectionSort(array):
    n = len(array)

    for i in range(n):
        max_index = i

        for j in range(i + 1, n):
            if array[j] > array[max_index]:
                max_index = j

        array[i], array[max_index] = array[max_index], array[i]

data = [12, 3, 8, 15, 5, 10]

selectionSort(data)
print("Sorted Array in Descending Order:")
print(data)
print("\nAmit_S057")
|
```

Output:

```
===== RESTART: C:/Users/mvlu
Sorted Array in Descending Order:
[15, 12, 10, 8, 5, 3]

Amit_S057
|
```
